

Deep Learning -based Anomaly Detection For Time-series Data In Industrial Control Systems

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Abstract: Industrial control systems(ICS) are widely used in key areas such as industry, energy, transportation, and water conservancy that can control the operation of production equipment. With the widespread use of sensors and actuators in ICS, these systems generate large amounts of multivariate time series data. Currently, there remain significant challenges in anomaly detection in ICS due to the lack of anomaly labels, high data volatility, and the demands of ultra-low inference times.

The paper proposes a deep learning-based multivariate anomaly detection method that combines Transform and GAN to detect anomalous behaviours in ICS. The method was tested on the SWaT and WADI datasets collected from industrial control systems, and experimental results show that the model proposed in this paper can outperform state-of-the-art baseline methods in anomaly diagnosis performance and timeliness.

Key Words: Anomaly Detection, Multivariate Time Series, GAN

1 Introduction

Industrial control systems(ICS) are widely used in critical areas such as industrial energy, transportation, and water conservancy to control the operation of production equipment. Typical ICS include programmable logic controllers, distributed control systems, and data acquisition and monitoring systems. Industrial control system security issues are directly related to the normalcy and sex of the national economy and social stability.

In many real-world systems (e.g., smart buildings, factories, power plants, and data centres), the proliferation of networked sensors and actuators generates large amounts of multivariate time-series data that can be used to continuously monitor the ICS'working conditions to detect anomalies in time so that the operators can take actions to investigate and resolve the underlying issues for these systems. Numerous ICS are created to carry out essential operations and are thus susceptible to being targeted by cyber-attacks, where attackers can exploit these vulnerabilities to infiltrate the system and trigger larger consequences, such as shutting down an entire production line or manipulating the production process. The study of attack detection methods for industrial control systems is of great importance to detect and stop hacker intrusion attacks in time and avoid causing physical damage.

In this paper, we propose a Transformer and GAN-based anomaly detection model(TGAD) to capture the dependencies on time series. The Anomaly-Attention[16] is used to capture the contextual sequences of subsequences in sliding windows to determine the long-term dependencies of subsequences in each window, and anomaly scoring

functions are constructed to detect anomalies at the end of the adversarial training phase.

The rest of the paper is structured as follows: Section 2 reviews the previous related work. Section 3 describes the design and analysis of the model. Section 4 presents the dataset, detailed information and experimental design, and evaluates the results. Finally, Section 5 concludes the paper and draws guidelines for future work.

2 Related Work

Previous research works were mainly based on modeling approaches to physical control systems [8,9,10,11,12]. Although research efforts are applying ML to anomaly detection in ICS [1,2,3,4,5,6,7], they rely on synthetic or data collected from information technology (IT) networks or small laboratory testbeds. The increasing number of sensor devices in the generation of IoT platforms also makes anomaly detection face two other challenges: lack of anomaly labels and high real-time speed requirements.

2.1 Classical methods

In previous work on ICS intrusion detection, principal component analysis (PCA) [8] and partial least squares (PLS) [9] are two popular unsupervised anomaly detection methods based on linear models. However, these models assume a Gaussian distribution in the data and are only valid for highly correlated data [10]. Angle-based outlier detection (ABOD) [11] and feature Bagging (FB) [12] are probabilistic models based on density estimation and are superior to distance-based methods. However, these methods do not apply to time series data because they do not consider time correlation [6].

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2.2 Deep Learning based methods

These problems have led researchers to develop countless unsupervised anomaly detection methods for automatic anomaly detection based on deep learning, such as the Lstm-ndt [1] method, which relies on deep LSTM-based neural network models, but as a recursive model, such models are slow to train in many cases where the input sequences are long. DAGMM [2] uses a deep self-coding Gaussian mixture model for dimensionality reduction of the feature space. However, it is inefficient and cannot exploit multimodal intercorrelation [3]. The Multiscale Convective Recursive Encoder-Decoder (MCRES) [4] converts the input sequence window into a normalized two-dimensional image. This method can capture more complex inter-modal correlations and temporal information but does not consider the case of insufficient training data. MAD-GAN [6] uses an LSTM-based GAN model to model time series distributions using generators that do not treat each data stream independently but consider the entire set of variables simultaneously in order to capture potential interactions between variables into the model, but the F1 score is still low. Instead of using resource-consuming recurrent models, recent models such as USAD [5] and GDN [3] use only attention-based network architectures to improve the training speed.

In this paper, we propose a multivariate time series anomaly detection model called TGAD of anomaly detection model for multivariate time series based on Transformer and Generative Adversarial Network. The parallel computation of the Transformer can meet the requirements of industrial control systems for real-time performance. The Transformer is embedded into the framework of GAN to solve the problem of lack of training data at each edge site, especially for multivariate and high-dimensional datasets, and to improve the speed and generation quality of traffic data generation.

3 The Method Of This Paper

We use insightful refactoring of the transformer architecture for the task of anomaly detection in time-series data. Just like other encoder-decoder models, in a transformer, an input sequence undergoes several attention-based transformations. This paper presents an anomaly detection architecture for time series data in industrial control systems (ICSs) that provides a fast and efficient learning model combined with parallel computing. Figure 1 shows the architecture of the neural network used in TGAD.

3.1 Problem Formulation

We consider the input as a sequence of data with a timestamp:

$$Z = \{z_1, z_2, \dots, z_T\} \quad (1)$$

where denotes the value of z_t in the time series at timestamp t . Since Z is a multivariate time series, each z is a vector of size m , i.e. the series has m features, which for the SWaT and WADI datasets is equal to the number of sensors and actuators. The model used in this paper is divided into two main parts, the training phase and the anomaly detection phase.

The training phase focuses on inputting a time series to predict anomalies (0 or 1) at the current moment, with 1 representing that the data point is anomalous. We try to detect the difference between an unknown sample and a normal sample by the anomaly score, which is then compared to a threshold to obtain the anomaly label, using only samples from normal data during training. To model the dependence between the current time point and the previous time point.

Given a training input time-series Z , for any unseen test time-series $\hat{Z}$ of length $\hat{T}$ and same modality as the training test time-series, we need to predict whether the datapoint at the t -th timestamp of the test set is anomalous. One GAN network, the generator (G), aims to generate realistic data, whereas the second one acts as a discriminator (D) trying to discriminate real data from that one generated by G. The training objective of G is to maximize the probability of D making a mistake, whereas the training objective D is to minimize its classification error. It is defined as:

$$G(x) = D(Z) \quad (2)$$

$$L = \|Z - D(G(x))\|_2 \quad (3)$$

and $\|\cdot\|_2$ denotes the L2-norm.

The encoder part takes the input x and maps it into a set of latent variables Z . In contrast, the decoder maps the latent variables Z back into the input space as a reconstruction. The difference between the original input vector Z and the reconstruction $\hat{Z}$ is called the reconstruction error L . Thus, the training objective aims to minimize this error.

In the anomaly detection phase, we will first predict the anomaly score for this window, calculate a threshold value using the previously entered anomaly score, and determine the occurrence of anomalies based on this threshold value.

3.2 The training phase

TGAD consists of two transformer encoders and two decoders. We consider the model inference in two phases. We first take as input x and Z (x is initially a zero matrix of Z dimensions), broadcast the relationship between the two distributions of the normal data Z learned and return, and continuously update the parameters of Encoder1 with the goal of generating a reconstruction that approximates the input window. Use the correlation differences computed in the multi-layer Anomaly Attention to compute the loss function:

$$L_1 = \|G(x) - Z\|_2^2 - \lambda \cdot \text{AssDis}(P, S; x) \quad (4)$$

$$L_2 = \|D(Z) - Z\|_2^2 - \lambda \cdot \text{AssDis}(P, S; Z) \quad (5)$$

We apply the position encoding and obtain the input of the first encoder, called I_1 , denoted by the first encoder performing the following operation:

$$I_1^1 = \text{LayerNorm}(\text{AnomalyAtte}(I_1) + I_1) \quad (6)$$

$$I_1^2 = \text{LayerNorm}(\text{FeedForward}(I_1^1) + I_1^1) \quad (7)$$

Applying the Anomaly-Attention proposed in [16] to Encoder2, which can model the prior-association and series-association simultaneously to embody the Association Discrepancy. To avoid the decoder from seeing future timestamp values during training, we adjust the self-attention in the window encoder to mask subsequent

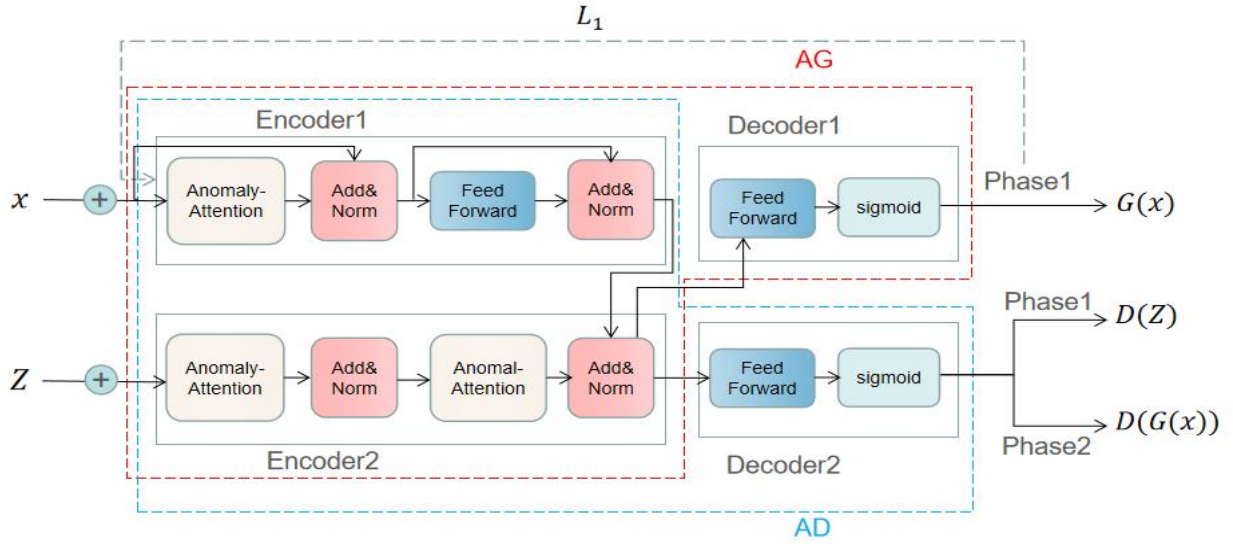


Fig. 1: Overall network framework

data positions. This modification is necessary because all data W is presented simultaneously for parallel training. We apply the position encoding to the input window Z to get I_2 and encoder Encoder2 performs the following operations:

$$I_2^1 = \text{Mask}(\text{AnomalyAtte}(I_2)) \quad (8)$$

$$I_2^2 = \text{LayerNorm}(I_2 + I_2^1) \quad (9)$$

$$I_2^3 = \text{LayerNorm}(\text{AnomalyAtte}(I_2^2) + I_2^1) \quad (10)$$

Since the complete input sequence up to the t th timestamp is used as input, it allows the model to encapsulate and exploit a larger context compared to the limited contexts available in the prior art. Finally, we perform the operation using two identical decoders:

$$G, D = \text{Sigmoid}(\text{FeedForward}(I_2^3)) \quad (11)$$

The activation function is used to generate an output in the range of $[0, 1]$, which matches the normalized input window.

The activation differences are smaller for anomalies, while for normal points, the correlation differences are larger. So the second stage is the adversarial process, wherein the generator work stage, we try to make the data generated by the generator learn the original data as well as possible, while the discriminator, on the other hand, needs to minimize the association difference as much as possible:

$$\min_{\text{Decoder1}} \max_{\text{Decoder2}} \|D(G(x)) - Z\|_2^2 - \lambda \cdot \text{AssDis}(P, S; Z) \quad (12)$$

λ is to weigh the loss condition, when $\lambda > 0$, the optimization is to expand the association difference and generate a normal sequence, the process of adversarial training can be decomposed as follows:

$$\min_{\text{Decoder1}} \|D(G(x)) - Z\|_2^2 - \lambda \cdot \text{AssDis}(P, S; Z) \quad (13)$$

$$\max_{\text{Decoder2}} \|D(G(x)) - Z\|_2^2 + \lambda \cdot \text{AssDis}(P, S; Z) \quad (14)$$

The training procedure of the model continues iteratively until the optimization algorithm reaches the convergence criteria based on the model loss. The dynamic evolution of the model loss in SWaT and WADI datasets during the training process is demonstrated in Figure 2:

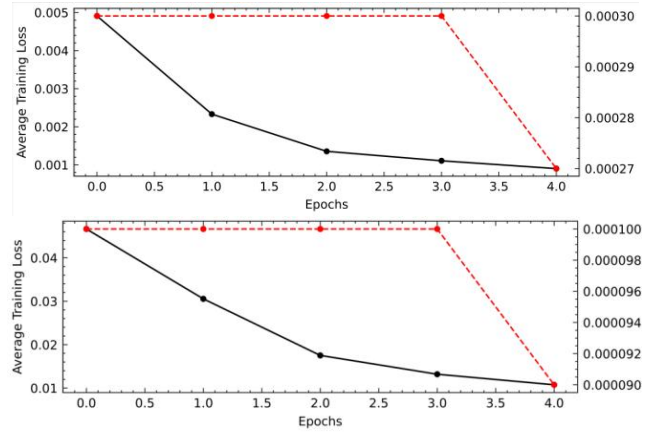


Fig. 2: The convergence process of the loss

3.3 The anomaly detection phase

Association-based anomaly criteria: We include normalized association differences in the reconstruction criteria, which benefit from both temporal representation and distinguishable association differences. When an outlier data point is reconstructed well, the association discrepancy tends to be larger, while if the reconstruction is poor, the reconstruction error tends to be larger. Thus, for unseen data, the final anomaly score is shown as follows:

$$S(\hat{Z}) = \text{softmax}(-\text{AssDis}(P, S; \hat{Z}))$$

$$\odot \quad (15)$$

$$\left[\frac{1}{2} \|D(G(\hat{Z})) - \hat{Z}\|_2^2 + \frac{1}{2} \|(G(\hat{Z})) - \hat{Z}\|_2^2 \right]$$

where $\odot$ is the element-wise multiplication. $S(\hat{Z})$ is the point-wise anomaly criterion of $\hat{Z}$. Typically, anomalies lead to a decrease in association discrepancy, which can result in a higher anomaly score. To overcome this challenge and improve anomaly detection performance, a design strategy can be employed to leverage the collaboration between reconstruction error and association discrepancy, ultimately improving the accuracy of anomaly detection. Based on the analysis, we designed a criterion to determine anomalies, which takes into account both

correlation differences and reconstruction errors, and can more accurately detect anomalous data.

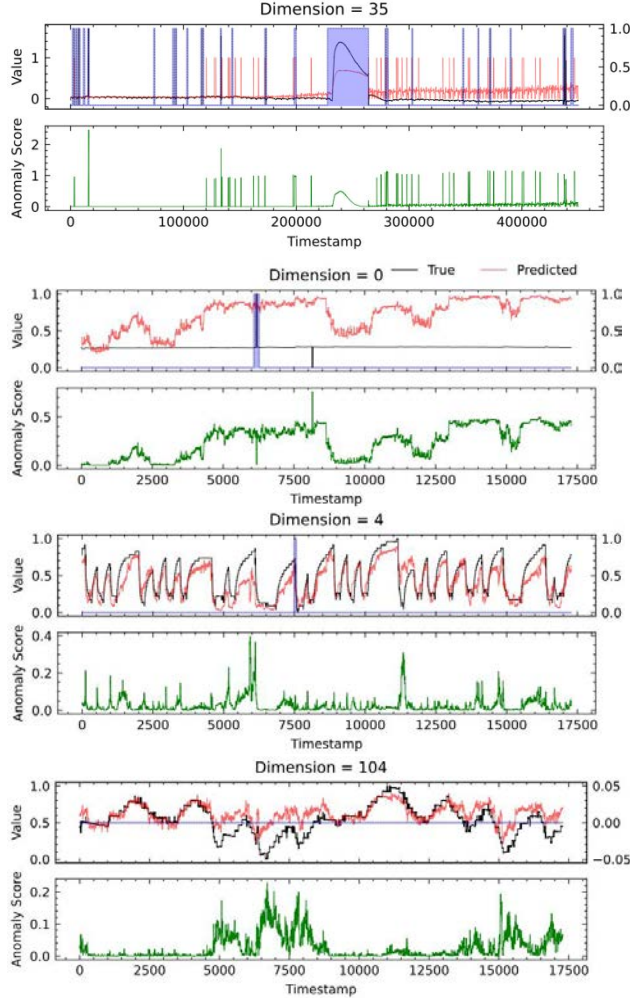


Fig. 3: Visualization of partial anomaly prediction:SWaT and WADI(black:true time series,red:presicted series,green:anomaly scores,blue:presicted anomaly)

The visualization of the partial anomaly prediction is shown in Figure 3. We label the current timestamp anomalous if any of the dimensions is anomalous. Figure 3 illustrates this process for a sample time-series.

4 Experiment Design

4.1 Datasets

Real datasets with anomaly labels are scarce, especially for large plants, and in this paper, we choose to record these as ground truth anomalies based on two sensor datasets from the physical system of a water treatment site: SWaT[13] and WADI[15], where operators simulate real-world water treatment plant attack scenarios. We initially explain our detection approach using the plant automation dataset and then compare the results with other datasets and other studies.

The SWaT/WADI data collection process lasted for 11/16 days with the systems operated 24 hours per day. Various cyber-attacks were executed on the test beds with different intents and divergent lasting durations (from a few minutes to an hour) in the final 4/2 days. The systems were either allowed to reach their normal operating state before another attack was launched or the attacks were launched

consecutively. Some general information about these two datasets is summarized in Table 1.

Table 1: Page Margins

Datasets	Features	Attacks	Train	Test	Anomalies
SWaT [13]	51	36	495 000	4499 19	11.97%
WADI [15]	127	15	1187 950	17281	5.99%

4.2 Data Preparation and System Architecture

For both of these datasets, we eliminated the first 21,600 samples from the training data (normal data) since it took 5-6 hours to reach stabilization when the system was first turned on according to [14].

In the anomaly detection process, we subdivide the original long multiple sequences into smaller time series by taking a sliding window across raw streams. We tried a set of different window sizes to capture the system status at different resolutions, namely $s_w = 30 \times i$, $i = 1, 2, \dots, 10$. To capture the relevant dynamics of SWaT data, the window is applied to the normal and testing datasets with shift length $s_s=10$.

4.3 Evaluation Metrics

We use the standard metrics, namely Precision (Pre), Recall (Rec), and F_1 scores, to evaluate the anomaly detection performance of the method in this paper:

$$\text{Pre} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (16)$$

$$\text{Rec} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (17)$$

$$F_1 = 2 \times \frac{\text{Pre} \times \text{Rec}}{\text{Pre} + \text{Rec}} \quad (18)$$

where TP is the correctly detected anomaly, FP is the falsely detected anomaly, TN is the correctly assigned normal, and FN is the falsely assigned normal.

4.4 Results

Table 2 provides the precision, recall and F1 for TGAD and baseline models for two datasets.

Table 2: Performance comparison of TGAD with baseline methods on the datasets.

	SWaT			WADI		
Method	Pre	Rec	F1	Pre	Rec	F1
PCA[8]	24.92	21.63	0.23	19.53	5.63	0.10
MAD-GAN[6]	95.93	63.74	0.77	22.33	33.92	0.37
USAD[5]	99.77	68.79	0.81	18.75	41.89	0.41
GDN[3]	96.97	69.67	0.81	29.12	40.19	0.42
TGAD[ours]	97.64	69.97	0.81	39.52	41.92	0.49

We extensively compare TGAD with four baseline models, including PCA [8], MAD-GAN [6], USAD [5], and GDN [3], for multivariate time-series anomaly detection using state-of-the-art models. Quantitative results for TGAD in two real-world datasets. The Pre, Rec and F1 represent the precision, recall and F1-score (as %)

respectively. F1-score is the harmonic mean of precision and recall. TGAD outperforms the baselines (in terms of Rec and F1score) for two datasets when we consider the complete dataset.

Table 3: Comparison of training times in seconds per epoch.

Method	SWaT	WADI
MAD-GAN[6]	27.79	297.60
USAD[5]	24.76	242.86
GDN[3]	59.41	4073.05
TGAD[ours]	18.34	173.91

Table 3 shows the average training times of the detection model associated with the generative adversarial network on two datasets in seconds per epoch. GDN does not use the generative adversarial network, and has good performance to learn rich feature representations of ICS, but it usually requires sufficient anomaly samples, and depends heavily on the quality of the training samples. It also has a high time complexity. For comparison, we report the training time for TGAD as the time it takes to discover discords in the test data. The training time of TGAD is 44%–75% lower than those of the baseline methods. This clearly shows the advantage of having a transformer model with positional encoding to push the complete sequence as an input instead of sequentially inferring over local windows.

5 Conclusion

In this paper, we evaluate the anomaly detection performance of our proposed model on two datasets. As mentioned earlier, we feed subsequences into our model. To compare our model's anomaly detection performance with other models, we conduct experiments on the same dataset. Additionally, we compare our model's training time with other GAN-based methods and obtain better results. This demonstrates that our proposed model has a faster response speed and is better adapted to industrial control systems. In future work, we will consider reducing the computational load by reducing the original dimensionality through PCA and building a lightweight model.

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